TOXICITY AND CARCINOGENICITY STUDIES OF 4-METHYLIMIDAZOLE IN F344/N RATS AND B6C3F1 MICE

4-Methylimidazole (4MI) is used in the manufacture of pharmaceuticals, photographic chemicals, dyes and pigments, cleaning and agricultural chemicals, and rubber. It has been identified as a by-product of fermentation in foods and has been detected in mainstream and side stream tobacco smoke. 4MI was studied because of its high potential for human exposure. Groups of 50 male and 50 female F344/N rats were fed diets containing 0-, 625-, 1,250-, or 2,500-ppm 4MI (males) or 0-, 1,250-, 2,500-, or 5,000-ppm 4MI (females) for 106 weeks. Based on the food consumption the calculated average daily doses were approximately 30, 55, or 115 mg 4MI/kg body weight to males and 60, 120, or 250 mg 4MI/kg to females. Survival of all exposed groups of males and females was similar to that of the control groups. Mean body weights of males in the 1,250- and 2,500-ppm groups and females in the 2,500- and 5,000-ppm groups were less than those of the control groups throughout the study. Feed consumption by 5,000-ppm females was less than that by the controls. Clonic seizures, excitability, hyperactivity, and impaired gait were observed primarily in 2,500- and 5,000-ppm females. The incidence of mononuclear cell leukemia in the 5,000-ppm females was significantly greater than that in the controls. The incidences of hepatic histiocytosis, chronic inflammation, and focal fatty change were significantly increased in all exposed groups of male and female rats. The incidences of hepatocellular eosinophilic and mixed cell foci were significantly increased in 2,500-ppm males and 5,000-ppm females. Groups of 50 male and 50 female B6C3F1 mice were fed diets containing 0-, 312-, 625-, or 1,250-ppm 4MI for 106 weeks. Based on the food consumption the calculated average daily doses were approximately 40, 80, or 170 mg 4MI/kg body weight to males and females. Survival of all exposed groups of males and females was similar to that of the control groups. Mean body weights of males and females in the 1,250-ppm groups and that in the 312- and 625-ppm females were less than those of the control groups. Feed consumption by exposed groups of male and female mice was similar to that by the controls. The incidences of alveolar/bronchiolar adenoma in all exposed groups of females, alveolar/bronchiolar carcinoma in 1,250-ppm males, and alveolar/bronchiolar adenoma or carcinoma (combined) in 1,250-ppm males and 625- and 1,250-ppm females were significantly greater than those in the control groups. The incidence of alveolar epithelial hyperplasia was significantly increased in the 1,250-ppm females. 4MI is carcinogenic inducing alveolar/bronchiolar adenoma and carcinoma in male and female mice. 4MI may also induce mononuclear cell leukemia in female rats.